Introduction

Over the past twelve weeks, I have embarked on the fifth module of my degree, Machine Learning (ML). The aim of the module was to gain a strong understanding of the concepts, theories, and tools used to develop ML models. This includes areas such as correlation, linear and polynomial regression, logistic regression, exploratory data analysis (EDA), model identification and evaluation. It was also an opportunity to reflect on the legal, ethical, and wider societal implications that must be considered when working with data and ML models in industry.

In this personal reflection, I will review and reflect on how my theoretical understanding of ML has grown, as well as my practical developmental capabilities in Python, including the use of different libraries to construct different types of models depending on the characteristics of the dataset.

All the activities I have completed in this module can be found in my ePortfolio linked here: https://liawilliams.github.io/machine-learning.html.

Development of Machine Learning (ML) Skills

Early in the module, I completed an exercise on EDA. In unit 2 I completed an EDA exercise, exploring a dataset featuring the fuel consumption of different cars (auto_mpg). I used several Python libraries including pandas, NumPy, matplotlib, and SciPy to retrieve key details about the data, clean and wrangle the data, view correlations between numeric variables, and encode categorical features ready for model development.

My primary challenge stemmed from my limited self-belief in completing EDA. I had performed EDA on several datasets in previous modules, however I wasn't confident in my abilities undertaking it, especially when using Python. Now I feel less apprehensive and nervous when conducting EDA and realised that developing strong skills in Python comes with persistent and repeated practise. This is reinforced by learning theory as practical experience using a particular skillset improves self-confidence during subsequent attempts (Reese, 2011).

My favourite activity was the unit 11 assignment, where I was tasked with developing a recommender system for a catalogue of songs hosted on Spotify. It was my first attempt at developing a model end-to-end, from initial data exploration and EDA through to evaluation. The assignment reintroduced me to scikit-learn for regression models, and to TensorFlow for deep learning (DL) models.

Once again, my main challenge was my lack of confidence when identifying which ML models would be the most suitable ones for my scenario. I used my knowledge which I gained in the previous units to evaluate which classical ML and DL models would be appropriate, settling on Random Forest (RF) and Multi-Layer Perceptron (MLP) models. I also considered wider deployment implications, researching different APIs and cloud systems which could be used to create a seamless pipeline, in addition to the data protection protocols and ethics associated with collecting real, behavioural data from individuals. These are all skills that I will use as a Data Scientist in industry.

Group Work and Collaboration with Others

For my unit 6 assignment, I was placed in a group of six and tasked with simulating a development team in Airbnb to apply regression and clustering techniques to determine which variables influence the price of a listing in New York. Professionally I have experience as a Project Manager within IT and technology, and I leaned on this experience to ask the team for regular progress updates during meetings and a group chat to ensure that we were on track to submit the written report by the deadline. I made sure that every team member assumed responsibility for certain tasks.

One of the key benefits of working in a group is exchanging ideas and perspectives that we would not have considered otherwise, such as employing K-Means segmentation to group listings into different market groups. It was also beneficial as at the time of completion, I was not confident in my coding abilities, so it meant that my colleagues could focus on the technical model development while I focused on writing sections of the report. I did experience feelings of frustration as certain parts of the report were not completed the standard that I would have completed it to, and this was reflected in the feedback received after marking. Nevertheless, the technical development and report were completed and submitted on time to a pass standard.

Several seminars were held throughout the module, presenting another opportunity to exchange ideas in real time with my course mates. It was tricky to attend all the live sessions due to existing work commitments, however the unit 4 seminar stands out to me, as I was able to witness a live demonstration of how scikit-learn is used to construct linear regression models between the population of a country and its GDP. I collaborated with my peers to identify what the target variable would be, GDP, and discussed when a linear regression model would be chosen as opposed to a classification model, such as decision trees.

Laal and Ghodsi (2012) identified that group work develops critical thinking capabilities more effectively than working independently, as exchanging ideas encourages participants to explain the logic and reasoning behind them. I experienced this in real time when participating in the seminars and during the group, as well as being able to share tips on weaker areas we possessed (such as Python).

Impact on Personal & Professional Development

Prior to undertaking this module, my knowledge of the different ML models which exist was limited to solely simple linear regression ones since that is what I had covered during previous modules. Terms such as Artificial Neural Networks (ANNs), clustering, and Python libraries like TensorFlow were unfamiliar to me, and I had no experience developing models end-to-end. Activities completed during the module, such as the assignments in unit 6 and 11, the perceptron activities in unit 7, and the Convolutional Neural Network (CNN) object recognition demo in unit 9 built my confidence and skillset. This will be crucial when I embark on my MSc Computing project in the coming months.

After the conclusion of the module, I will focus on deepening my ML capabilities and expand my knowledge of the application of different types of models to various categories of data. I will also incorporate skills gained in previous modules, such as web scraping to retrieve data, to complete more complex ML projects to become proficient in model development, to make me a competent and skilled Data Science professional.

References

Laal, M. and Ghodsi, S.M. (2012) 'Benefits of collaborative learning', Procedia - Social and Behavioral Sciences, 31, pp. 486–490. Available at: https://doi.org/10.1016/j.sbspro.2011.12.091.

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